FIBONACCI RELATIONS

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Abstract. Following Melhain [7] , Cheng and Mong [13] give a list of 12 identities involving Fibonacci numbers, which they attribute to various authors. We give a diagramatic interpretation (Figure 1) of these using λ-lengths of arcs on a hyperbolic annulus , and then we show how to derive them using essentially Penner's Ptolemy relation [8].

As always we claim no originality, and we are sure that this is well known to experts. It's merely exercise in organisation and evaluating the limits of geometric intuition.

1. Introduction

The Fibonacci numbers are defined by the recurrence relation with the first two terms given by $F_0 = 0$ and $F_1 = 1$:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \ldots$$

There is a closed formula for F_n called the Binet formula, which remarkable as it is we shall not need here:

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right). \tag{1}$$

Integers are usually represented on a number line, but we will represent the Fibonacci numbers as vertices of a regular graph as in Figure 1. We will see how properties of the Fibonacci numbers are reflected in the geometry of this graph. The graph naturally partitions the Fibonacci numbers into two sets, the even indexed Fibonacci numbers and the odd indexed Fibonacci numbers.

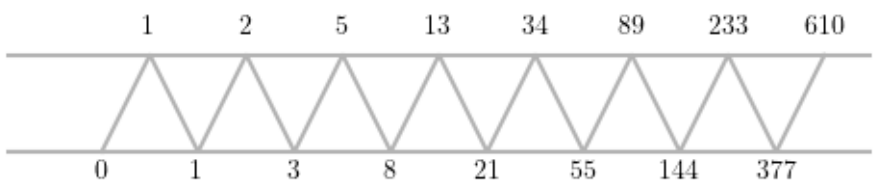

Figure 1. Fibonacci numbers as vertices of a graph: odd index Fibonacci numbers are above and even below.

1.1. **Identities.**

$$F_{n+1} = F_n + F_{n-1}, \tag{2}$$

but they also satisfy many other identities. We will be concerned with these and the relationship between them. In particular, we will be concerned with the following two identities:

$$F_{n-1}F_{n+1} - F_n^2 = (-1)^n \tag{3}$$

and

$$F_{n-2}F_{n-1}F_{n+1}F_{n+2} - F_n^4 = -1 \tag{4}$$

According to Dickson [2], pages 393 and 401, the first of these identities was proved by Robert Simson and the second was stated by E. Gelin and proved by E. Cesàro. Simson's identity and the Gelin-Cesàro identity have been generalized many times.

The identity (3)is an immediate consequence of the matrix identity:

$$\begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n \tag{5}$$

and the fact that

$$\det \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = -1.$$

In fact, all the λ-lengths we consider are simply determinants of 2×2 matrices whose entries are Fibonacci numbers. It will be convenient to extend the definition of Fibonacci numbers to negative integers by using the matrix identity (5) above. The F_n will be the lower left (= upper right) hand entry of the nth power of our matrix on the left so that in particular $F_0 = 0$ and $F_{-1} = 1$ and more generally

$$F_{-n} = (-1)^{n+1} F_n.$$

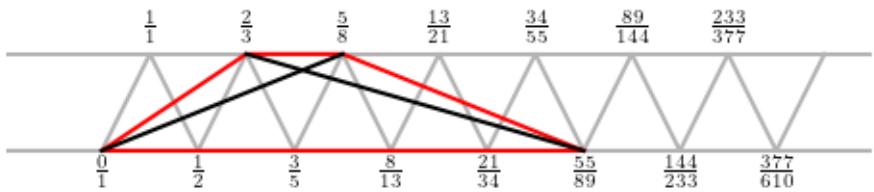

FIGURE 2. Fibonacci fractions

1.2. **History according to Hendel.** Hendel gives a brief account of development of the identities (3) and (4) and their generalizations:

$$F_{n+1}F_{n-1} = F_n^2 + (-1)^n, \tag{6}$$

$$F_{n+a}F_{n-a} = F_n^2 + (-1)^{n+a+1}F_a^2, \tag{7}$$

$$F_{n+a}F_{n+b} = F_nF_{n+a+b} + (-1)^nF_aF_b. \tag{8}$$

aredue to Cassini, Catalan and Tagiuri respectively (see [3] and [4] for details).

1.3. **Melhain new identity.** They prove the following variant of the Gelin-Cesàro identity:

$$F_{n+1}F_{n+2}F_{n+6} - F_{n+3}^3 = -1^nF_n \tag{9}$$

In [1] Cheng and Mong propose a method of proof for this kind of identity:

- show both sides satisfy the same recurrence relation
- check the identity for enough initial values to guarantee the identity holds for all n.

1.4. **ChatGPT text conversion of Melhain and Shannon's introduction.** One method of proof is to use the Cassini identity (3) to substitute for $(-1)^n$, then express each of F_{n-1}, F_n, F_{n+3}, and F_{n+6} in terms of F_{n+1} and F_{n+2}, and expand both sides. We prefer this method of proof since it carries over nicely to our generalization of (3), which we give next.

Our generalization is stated for the sequence $\{W_n\} = \{W_n(a, b; p, q)\}$ defined by

$$W_n = pW_{n-1} - qW_{n-2}, \quad W_0 = a, \quad W_1 = b$$

where a, b, p, q are taken to be arbitrary complex numbers with $q \neq 0$. Since $q \neq 0$, $\{W_n\}$ is defined for all integers n. Put $e = pab - qa^2 - b^2$. Then

$$W_{n+1}W_{n+2}W_{n+6} - W_{n+3}^3 = eq^{n+1}(p^3W_{n+2} - q^2W_{n+1}) \tag{4}$$

2. Chen and Mong's list

Chen and Mong give a list of identities having done a search of the literature in particular they cite Long [5], and Weisstein [10]. Below is an abridged list omitting those entries which involve Lucas numbers and those which are proven using induction.

(c1)

$$F_{n+a}F_{n+b} - F_nF_{n+a+b} = (-1)^nF_aF_b$$

(c4)

$$F_{n-1}F_{n+1} - F_n^2 = (-1)^{n-1}$$

(d4)
$$F_n^2 - F_{n-2}F_{n+2} = (-1)^n$$
(d6)
$$F_n^2 - F_{n-1}F_{n+1} = (-1)^{n-1}$$
(c11)
$$F_n^2 + (-1)^{n+r-1}F_r^2 = F_{n-r}F_{n+r}$$
(d8)
$$F_n^2 - F_{n-1}^2 = F_nF_{n-1} + (-1)^{n-1}$$
(c7)
$$F_{n+m} = F_{n-1}F_m + F_nF_{m+1}$$
(d1)
$$F_{2n} = F_{n+1}^2 - F_{n-1}^2$$
(d2)
$$F_{2n+1} = F_{n+1}^2 + F_n^2$$
(d9)
$$F_{2n+1} + (-1)^n = F_{n-1}F_{n+1} + F_{n+1}^2$$
(d12)
$$F_{2n+1} = F_{n+3}F_n - F_{n+1}F_{n-1}$$
(d3)
$$F_{n+2}F_{n-1} = F_{n+1}^2 - F_n^2$$
(d7)
$$F_nF_{n+3} = F_{n+1}F_{n+2} + (-1)^{n-1}$$
(c11)
$$F_n^2 + (-1)^{n+r-1}F_r^2 = F_{n-r}F_{n+r}$$
(c2)
$$F_{n+1}^2 = 4F_nF_{n-1} + F_{n-2}^2$$
(c9)
$$F_{2k+1}F_{2n+1} = F_{n+k+1}^2 + F_{n-k}^2$$
(c12)
$$F_n^4 - F_{n-2}F_{n-1}F_{n+1}F_{n+2} = 1$$

3. Proofs

We'll give proofs grouping identities as per the previous section. The identity (c12) is just a restatement of the Gelin-Cesàro identity (4) and we'll leave this till the end.

3.1. **Variations on Cassini: (c1), (c4), (d4), (d6), (c11),(d8).** Let's begin with the four identities. The first on the list (c1) is called Tagiuri's identity see also discussion `https://math.stackexchange.com/questions/1356391/is-there-a-name-for-this-fibonacci-identity` for Vajda's identity.

$$0 = (a-b)(a+b) - (a-c)(a+c) + (b-c)(b+c)$$
$$0 = (a-b)(a+b) + (b-c)(b+c) + (c-a)(c+a)$$

Somos `https://grail.eecs.csuohio.edu/%7Esomos/ident04.gp`

$$\begin{aligned} F_{n+a}F_{n+b} - F_nF_{n+a+b} &= (-1)^n F_aF_b \\ F_{n-1}F_{n+1} - F_n^2 &= (-1)^{n-1} \\ F_n^2 - F_{n-2}F_{n+2} &= (-1)^n \\ F_n^2 - F_{n-1}F_{n+1} &= (-1)^{n-1} \end{aligned}$$

The last identity is just a restatement of the second (with a La dernière identité n'est qu'une reformulation de la seconde (avec un possible missprint). The the first is due to Tagiuri according to Hendel, the second (Cassini's identity) and the third are special cases of the first where $a = 1, b = -1$ and $a = 2, b = -2$ respectively.

Another, perhaps more appropriate way is to say that (c4) is a generalization of Cassini's identity.

Not also that, modulo a possible missprint, (c11) is of this form:

$$F_{n-r}F_{n+r} - F_n^2 = (-1)^{n+r-1}F_r^2$$
$$F_{n-r}F_{n+r} - F_n^2 = (-1)^n F_r^2$$

Finally, (d8) which says:

$$F_n^2 - F_{n-1}^2 = F_nF_{n-1} + (-1)^{n-1}$$

is a bit of a "joker" since rearranging one has:

$$F_n^2 - F_{n-1}(F_{n-1} + F_n) = F_n^2 - F_{n-1}F_{n+1} = (-1)^{n-1}.$$

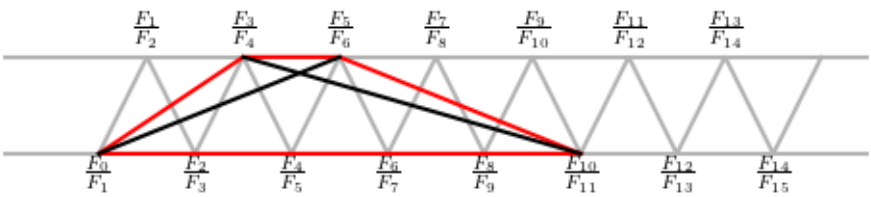

FIGURE 3. Fibonacci numbers

3.2. **Matrix multiplication (c7), (d1), (d2).**

$$\begin{aligned} F_{n+m} &= F_{n-1}F_m + F_nF_{m+1} \\ F_{2n} &= F_{n+1}^2 - F_{n-1}^2 \\ F_{2n+1} &= F_{n+1}^2 + F_n^2 \end{aligned}$$

Observe that the second is a consequence of the first by setting $m = n$ (and using the recurrence to replace for F_n). The third is also a consequence of the second by setting n to $n+1$ and using the recurrence relation twice:

$$\begin{aligned} F_{2n+1} &= F_{n-1}F_{n+1} + F_nF_{n+2} \\ &= F_{n-1}F_{n+1} + F_n(F_{n+1} + F_n) \\ &= F_{n-1}F_{n+1} + F_nF_{n+1} + F_n^2 \\ &= F_{n+1}^2 + F_n^2. \end{aligned}$$

In fact (C7) is a consequence of the matrix identity (5) since:

$$\begin{pmatrix} F_{n+m+1} & F_{m+n} \\ F_{m+n} & F_{m+n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{m+n} = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \begin{pmatrix} F_{m+1} & F_m \\ F_m & F_{m-1} \end{pmatrix}$$

Clearly the value of the lower left hand coefficient is F_{m+n} and computing it on the right hand side gives the desired identity.

3.3. **Norms in $\mathbb{Z}[i]$ and c9.**

$$F_{2k+1}F_{2n+1} = F_{n+k+1}^2 + F_{n-k}^2$$

$$F_{2n+1} = F_{n+1}^2 + F_n^2 = |F_{n+1} + iF_n|^2$$

3.4. **Factorisation: Gelin-Cesàro (c12).** The identity (c12) is just a restatement of the Gelin-Cesàro identity:

$$F_{n-2}F_{n-1}F_{n+1}F_{n+2} - F_n^4 = -1 \tag{10}$$

Rearranging and factorising one has the more symmetric form:

$$(F_{n-2}F_{n+2})(F_{n-1}F_{n+1)}) = F_n^4 - 1 = (F_n^2 - 1)(F_n^2 + 1).$$

Now using the Cassini identity yields:

$$\begin{aligned} F_{n-2}F_{n+2} &= (F_n^2 + (-1)^n F_2) \\ F_{n-1}F_{n+1} &= (F_n^2 + (-1)^n F_1) \end{aligned}$$

then note $F_1 = F_2 = 1$ so the result follows.

4. Recurrences for powers etc

An general rule of the theory of linear recurrences, and this is in some sense what Cheng and Mong are exploiting in [1], is that powers of a linear recurrence also satisfy a linear recurrence. For example, the Fibonacci numbers satisfy the recurrence (2) and their squares satisfy:

$$F_{n+1}^2 = 2F_n^2 + 2F_{n-1}^2 - F_{n-2}^2 \tag{11}$$

More formally: if a sequence x_n has a characteristic equation with roots α and β, then the sequence of its squares x_n^2 will have a characteristic equation whose roots are α^2, β^2, and $\alpha\beta$.For the Fibonacci sequence, the roots of the characteristic equation ($x^2 - x - 1 = 0$) are:

- $\phi = \frac{1+\sqrt{5}}{2}$ (The Golden Ratio)
- $\psi = \frac{1-\sqrt{5}}{2}$ (The Conjugate of the Golden Ratio)
- $\Rightarrow \phi\psi = -1$

ans so one can compute the characteristic equation for the squares of the Fibonacci numbers as follows:

$$(x - \phi^2)(x - \psi^2)(x - \phi\psi) = x^3 - 2x^2 - 2x - 1.$$

The question is how to visualise this recurrence relation for the squares of the Fibonacci numbers.

$$F_{n+1}^2 - F_n^2 = (F_n^2 + F_{n-1}^2) + (F_{n-1}^2 - F_{n-2}^2)$$

$$\begin{aligned}
\text{LHS} &= F_{n+1}^2 - F_n^2 \\
&= (F_{n+1} - F_n)(F_{n+1} + F_n) \\
&= F_{n-1}(F_{n+1} + F_n) \\
&= F_{n-1}F_{n+2}
\end{aligned}$$

$$\begin{aligned}
\text{RHS} &= (F_n^2 + F_{n-1}^2) + (F_{n-1}^2 - F_{n-2}^2) \\
&= F_{2n-1} + (F_{n-1} - F_{n-2})(F_{n-1} + F_{n-2}) \\
&= F_{2n-1} + F_{n-3}F_n
\end{aligned}$$

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